DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY
UNIVERSITY EXAMINATIONS 2021/2022 ACADEMIC YEAR
THIRD YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

CIT 3203: RESEARCH METHODS

DATE: 8/12/2021 **TIME: 08.30-10.30 A.M.**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

(a) State the relationship between:
   i. Research and Science (2 marks)
   ii. Research and theory (2 marks)

(b) State the difference between a sample and a population (2 marks)
(c) Define hypothesis in research (2 marks

(d) In your area of specialization, choose a research article or a thesis and identify the following:
   i. The problem statement (1 mark)
   ii. The purpose statement (1 mark)
   iii. Objectives (1 mark)
   iv. hypothesis (1 mark)

(e) List three characteristics of a good hypothesis (3 marks)

(f) You have been hired as a research assistant in an NGO firm. List four research ethics you are required to uphold during execution of your duties. (4 marks)

(g) Differentiate the following sampling techniques:
   i. Probability (2 marks)
   ii. Biased sampling (2 marks)

(h) List three types of biased sampling (3 marks)

(i) Explain the following measurement scales:
   i. Nominal (2 marks)
   ii. Ordinal (2 marks)

Question 2 :(20 Marks)

(a) Outline three primary data collection techniques. (3 marks)

(b) Highlight the seven steps involved in a research process. (7 marks)

(c) Differentiate between structured interviews and unstructured interviews. (4 marks)

(d) Discuss six advantages of conducting interviews as an instrument for research. (6 marks)

Question 3: (20 Marks)

a) Discuss the following types of research:

i. Descriptive vs Analytical (2 marks)
ii. Applied vs Fundamental (2 marks)

b) Makini University is conducting research on Alcohol and Drug abuse among the University staff. The data collected will be analyzed to determine the extent of alcohol and drug abuse.

Required

i). Explain a suitable research design (4marks)
ii). State four reasons for selecting the research design in (i) above (4 marks)

C) Explain four reasons for developing a hypothesis. (8 marks)

Question 4: (20 Marks)

a) IT class in Kaimosi college of agriculture conducted a study on insects to establish their effects on pollination.

Given the above information, required:

i. discuss three research ethics required in conducting the research. (6 marks)
ii. Explain four reasons for selecting observation rather than questionnaire in the data collection (4 marks)

b) Distinguish the following terms as used in research.

i. Theoretical framework and conceptual framework (2 marks)
ii. Null hypothesis and alternative hypothesis (2 marks)

c) Explain three purposes of a conceptual framework in research (6 marks)